

Throughput Diagnostics Report

Enterprise ML Analysis Output

Source CSV: d:/Throughput_Tool/throughput_20260423_180414.csv

Generated: 2026-05-08 23:57:15

Metric	Value
Total Rows Analyzed	581
Total Anomalies	581
Critical Rows	253
Warning Rows	328
Maximum Latency	1780.00 us
Average Latency	507.19 us
Health/Performance Score	78.24/100

Executive Summary

The ML pipeline analyzed **581** rows and detected **581** anomalous rows. Critical findings: **253**, warnings: **328**.

Row-wise and Column-wise Findings

Category	Count
Row-wise anomaly findings	0
Column-wise anomaly findings	436

Critical Frame Ranking

Rank	Frame	Peak Latency (us)
1	Frame_309	1780.00
2	Frame_435	1080.00
3	Frame_475	1080.00
4	Frame_175	1080.00
5	Frame_155	1080.00
6	Frame_195	1080.00
7	Frame_455	1080.00
8	Frame_275	1080.00
9	Frame_115	1080.00
10	Frame_235	1080.00

ML-generated Explanations

T=1 [Warning] Frame_53 WARNING SPIKED to 740us (avg 663us, delta 76us, jump 0us, z=0.58) | Frame_309 WARNING DROPPED to 600us (avg 753us, delta -153us, jump 0us, z=0.93)

T=2 [Warning] Frame_53 WARNING DROPPED to 600us (avg 663us, delta -63us, jump -140us, z=0.48) | Temporal instability: moving-average drift and rolling variance spike detected

T=3 [Warning] Frame_53 WARNING DROPPED to 580us (avg 663us, delta -83us, jump -20us, z=0.64) | Temporal instability: moving-average drift and rolling variance spike detected

T=4 [Warning] Temporal instability: moving-average drift and rolling variance spike detected

T=5 [Critical] Frame_53 CRITICAL SPIKED to 920us (avg 663us, delta 256us, jump 320us, z=1.95) | Temporal instability: moving-average drift and rolling variance spike detected

T=6 [Critical] Frame_53 CRITICAL DROPPED to 580us (avg 663us, delta -83us, jump -340us, z=0.64) | Temporal instability: moving-average drift and rolling variance spike detected

T=7 [Warning] Temporal instability: moving-average drift and rolling variance spike detected

T=8 [Warning] Temporal instability: moving-average drift and rolling variance spike detected

T=9 [Critical] Frame_53 CRITICAL SPIKED to 920us (avg 663us, delta 256us, jump 320us, z=1.95) | Temporal instability: moving-average drift and rolling variance spike detected

T=10 [Critical] Frame_53 CRITICAL DROPPED to 580us (avg 663us, delta -83us, jump -340us, z=0.64) | Temporal instability: moving-average drift and rolling variance spike detected

T=11 [Warning] Temporal instability: moving-average drift and rolling variance spike detected

T=12 [Warning] Frame_53 WARNING DROPPED to 580us (avg 663us, delta -83us, jump -20us, z=0.64) | Temporal instability: moving-average drift and rolling variance spike detected

Recommendations

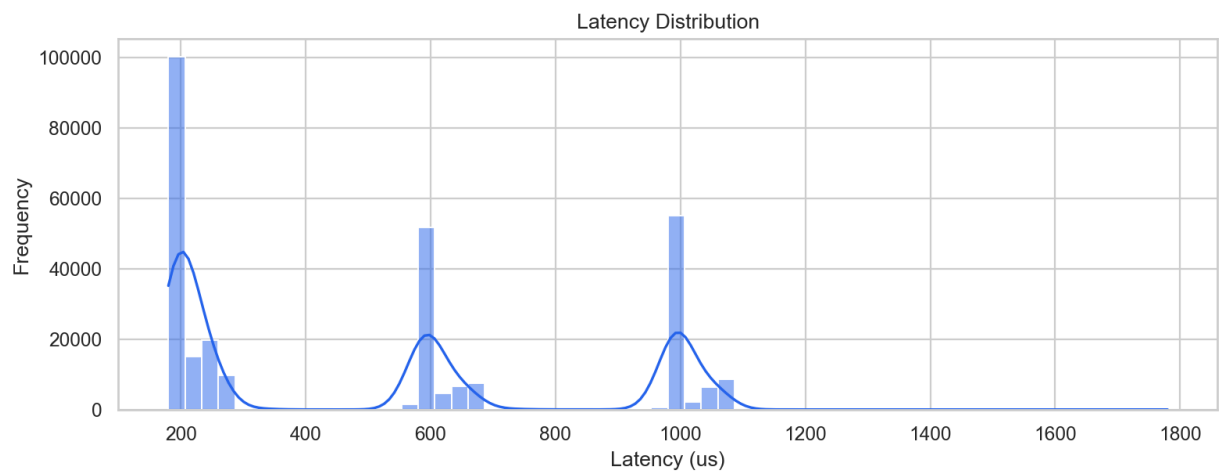
- Investigate high-variance frames and optimize scheduler timing for top-ranked critical frames.
- Review warning frames for early signs of instability and threshold drift.

Final Conclusion

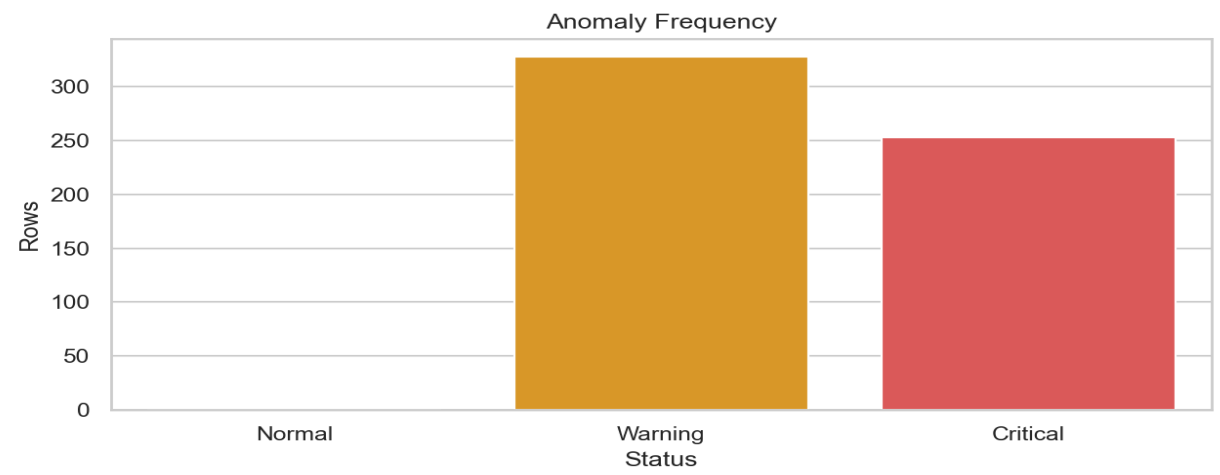
Conclusion: The diagnostics pipeline executed successfully and produced actionable insights with anomaly classification, frame criticality ranking, and performance health scoring.

Visual Charts

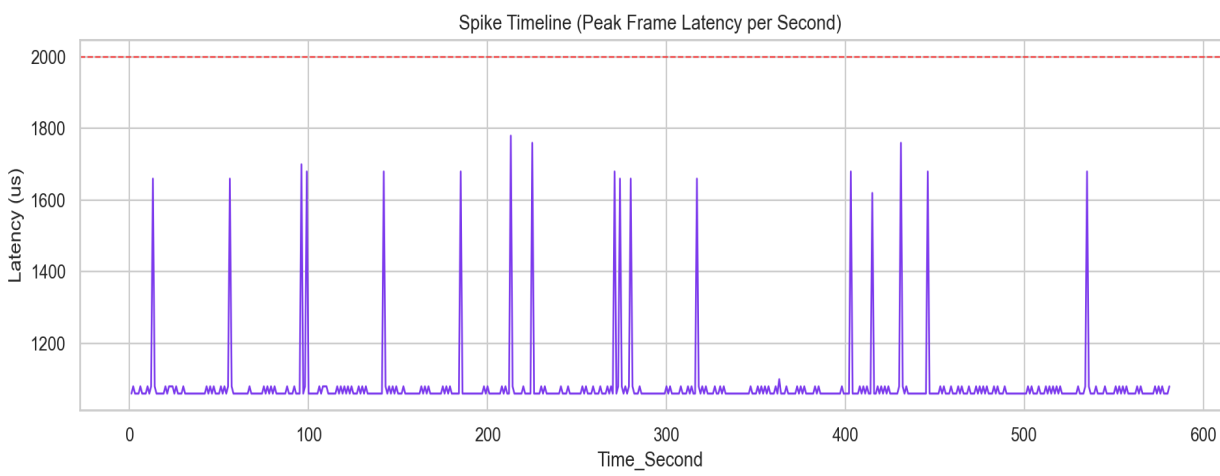
Latency Distribution



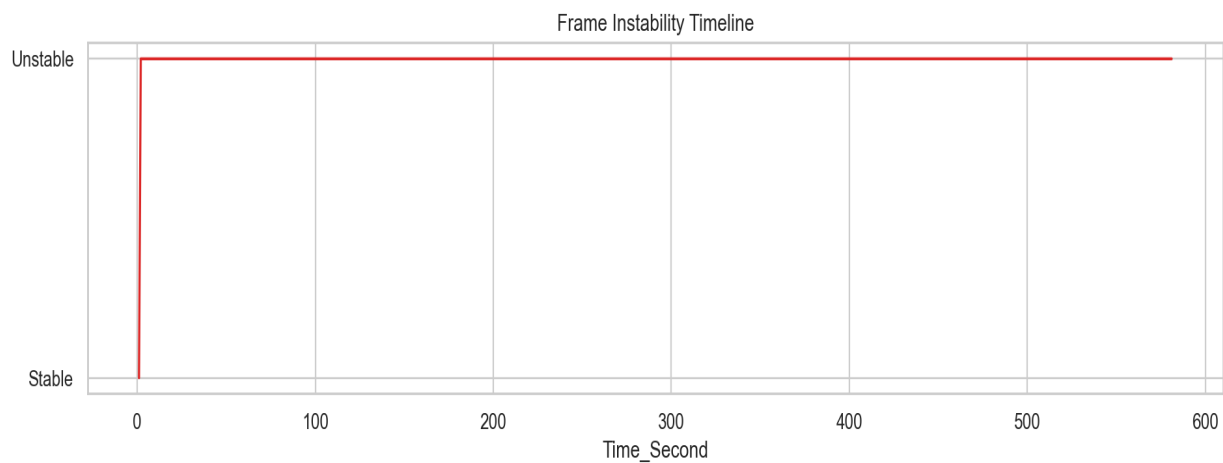
Anomaly Frequency



Spike Timeline



Frame Instability



Top Critical Frames

